SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	2024	**Date:**	26-08-26

Code of Conduct

*I ____***T Bhanu Teja/192472259***______ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____Bhanu Tejas__________

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

1. **Given the following paragraph:**
"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

Tasks: a) Identify all referring expressions (mentions) and possible antecedents.

b) Apply the following **constraints** to resolve the references:

- Gender and number agreement
- Recency (prefer recent antecedents)
- Semantic compatibility (e.g., animacy and action suitability)
- Coherence (maintain consistent entity chains)

c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

ANSWER:

a) Referring expressions and antecedents

Given:

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

Mention	Possible antecedent	Resolved antecedent
He	John, Mary	**John**
She	John, Mary	**Mary**
it	ball, park	**ball**
him	John, Mary, dog	**John**
they	John, Mary, dog	**John + Mary + dog**
all	John, Mary, dog	**John + Mary + dog**

b) Applying constraints

- **Gender and number:** "He" is masculine singular, so it refers to **John**. "She" refers to **Mary**.
- **Recency:** Recent compatible entities are preferred.
- **Semantic compatibility:** A ball is something suitable to play with, so **it → ball**.
- **Coherence:** The references should form consistent entity chains.

c) Constraint graph

Coreference Resolution

↓ ↓ ↓

Gender Recency Semantics

↓

Final References

↓ ↓ ↓

He → John She → Mary it → ball

↓

him → John

↓

they/all → John + Mary + Dog

d) Final resolved paragraph

John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home.

e) Constraint priority

A suitable order is:

Gender/Number → Semantic Compatibility → Recency → Coherence

If recency is relaxed, the system may consider older entities and therefore produce more ambiguous or incorrect references.

2. Constraint-Based Dialogue Planning

Given user input

"I have an important exam tomorrow but I'm not able to concentrate."

Required dialogue act: Advice + Encourage

a) Discourse planning steps

1. Identify the user's problem: **lack of concentration**.
2. Identify the context: **important exam tomorrow**.
3. Plan an **advice + encouragement** response.
4. Include at least two keywords: **focus, break, confident**.
5. Check length, coherence and positive tone.

Flowchart

User Input

↓

Identify Problem

↓

Exam + Lack of Concentration

↓

Advice + Encouragement

↓

Add Required Keywords

↓

Check Constraints

↓

Final Response

b) Three possible responses

Response 1:

Take a short break and then focus on one topic at a time. Stay confident and prepare calmly for tomorrow's exam.

Response 2:

Try studying in short sessions with small breaks to improve your focus. Stay confident and complete small goals before the exam.

Response 3:

Relax for a few minutes and then focus on the most important topics. Take regular breaks and stay confident about your preparation.

c) Best response

Response 2 is the best because it satisfies all constraints:

- Gives advice
- Encourages the student
- Uses **focus, break, confident**
- Contains only two sentences
- Maintains a positive tone
- Directly addresses the concentration problem

d) Effect of violating constraints

If the response is too long, it becomes difficult to follow. If the positive-tone constraint is violated, the chatbot may sound discouraging or judgmental, reducing user satisfaction.

3. Constraint-Based Word Sense Disambiguation

Given sentence

"The bank by the river flooded after the storm, but it was saved by quick action."

a) Resolve "bank"

The word **bank** has two possible meanings:

1. Financial institution
2. Riverbank

The phrase **"by the river"** and the fact that the bank **"flooded"** strongly indicate the **riverbank** meaning.

Therefore:

$$\boxed{bank = Riverbank}$$

Constraint analysis

Context clue	Result
"by the river"	Riverbank ✓
"flooded"	Riverbank ✓
Physical location	Riverbank ✓
Financial activity	Not present

b) Predicate logic representation

Let:

- RiverBank(x) = x is a riverbank
- River(y) = y is a river
- By(x,y) = x is beside y
- Flooded(x) = x flooded
- Storm(s) = s is a storm
- Saved(x,a) = x was saved by action a
- QuickAction(a) = a is a quick action

Representation:

$$\boxed{RiverBank(x) \wedge River(y) \wedge By(x,y) \wedge Flooded(x) \wedge Storm(s) \wedge After(F, loodingStorm) \wedge Saved(x,a) \wedge Q}$$

This represents the main entities and relationships in the sentence.

c) Target-language sentence

Simple English

The riverbank flooded after the storm, but quick action saved it.

Telugu

తుఫాను తర్వాత నది ఒడ్డు వరదకు గురైంది, కానీ వేగవంతమైన చర్యతో అది రక్షించబడింది.

d) Simple RST discourse tree

The word **“but”** indicates a **Contrast** relationship.

```
        ROOT

         |

      CONTRAST

     /      \

    /        \

Riverbank flooded    It was saved
```

after the storm   by quick action

Nucleus: Riverbank flooded after the storm
Satellite: It was saved by quick action

e) Advantages of constraint-based approach

1. **Reduces ambiguity** by using contextual constraints.
2. **Preserves important entities and relationships.**
3. **Improves semantic accuracy** in the final output.
4. **Provides interpretable decisions**, such as selecting riverbank because of "river" and "flooded."
5. It is useful in **healthcare, banking, customer support and other domain-specific NLP systems**.

Overall reference diagram

Input Sentence

↓

Context Analysis

↓

Identify Ambiguity

↓

Apply Constraints

↓

Semantic Interpretation

↓

Predicate Representation

↓

Coherent Output

These versions are **much more suitable to write in an assessment**: they cover every sub-question, show the required NLP concepts, and include diagrams without making the answers unnecessarily huge.

2. **Conversation History:** User: “I have an important exam tomorrow but I’m not able to concentrate.”

Required Dialog Act: Advise + Encourage (Give advice and motivate the student).

Constraints:

1. Maintain **entity coherence** (reuse “exam”, “concentrate”, “you”).
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

Tasks: a) Identify the relevant discourse planning steps needed before generation.

b) Generate **three** different possible responses that satisfy **all** the above constraints.

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

d) Explain how violating any two constraints would affect the quality of the dialog.

ANSWER:

a) Dialogue planning steps

The system first identifies the user's main problem and then plans a response according to the given constraints.

Step 1: Identify the problem

The user has:

- An important exam tomorrow.
- Difficulty concentrating.

Step 2: Identify the intention

The user is indirectly seeking **help and motivation**.

Therefore:

$$Intent = Advice + Encourage$$

Step 3: Identify important entities

Entity	Value
Event	Exam
Time	Tomorrow
Problem	Lack of concentration

Step 4: Select response content

The response should contain at least two of the required concepts:

- **Focus**
- **Break**
- **Confident**

Step 5: Apply constraints

The final response must:

- Contain **2–3 sentences**
- Give useful advice
- Encourage the user
- Maintain a positive and polite tone
- Maintain consistency with the exam context
- Include at least two required keywords

b) Dialogue planning flowchart

User Input

↓

Identify User Problem

↓

Exam + Lack of Focus

↓

Identify Intent

↓

Advice + Encouragement

↓

Select Response Content

↓

↓ ↓ ↓

Focus Break Confident

↓

Constraint Checking

↓

Length + Tone + Coherence

↓

Final Response

c) Possible responses

Response 1

“Take a short break and then focus on one topic at a time. Stay confident and prepare calmly for tomorrow’s exam.”

Response 2

“Try studying in short sessions with small breaks to improve your focus. Stay confident and complete small goals before the exam.”

Response 3

“Relax for a few minutes and then focus on the most important topics. Take regular breaks and stay confident about your preparation.”

---

d) Evaluate the responses

Constraint	**Response 1**	**Response 2**	**Response 3**
Advice	✓	✓	✓
Encouragement	✓	✓	✓
Focus	✓	✓	✓
Break	✓	✓	✓
Confident	✓	✓	✓
2–3 sentences	✓	✓	✓
Positive tone	✓	✓	✓

Best response

Response 2 is highly suitable because it directly addresses the concentration problem and provides a practical study strategy.

“Try studying in short sessions with small breaks to improve your focus. Stay confident and complete small goals before the exam.”

It satisfies the required **Advice + Encourage** dialogue act and maintains a positive and supportive tone.

e) Importance of constraints in dialogue planning

Constraints prevent a dialogue system from generating irrelevant or inappropriate responses. For example, a response that discusses unrelated topics would violate **entity coherence**. Similarly, a very long response would violate the **length constraint**.

A constraint-based dialogue system therefore checks:

$$\text{Intent} + \text{Content} + \text{Length} + \text{Tone} + \text{Coherence}$$

before generating the final response.

Conclusion

Constraint-based dialogue planning allows a chatbot to generate responses that are **relevant, concise, coherent, polite, and goal-oriented**. In this example, the system understands that the user needs both practical study advice and encouragement, while ensuring that the final response satisfies all specified constraints.

3. **Source Sentence:** “The bank by the river flooded after the storm, but it was saved by quick action.”

Note: “bank” is ambiguous (financial institution or riverbank).

Constraints:

1. Correct **Word Sense Disambiguation** using context.
2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

Tasks: a) Resolve the ambiguous word “bank” and justify using constraints.

b) Write a predicate logic representation of the sentence.

c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

ANSWER:

a) Resolve the ambiguity of the word “bank”

Possible senses

1. **Financial bank** — an organization dealing with money and accounts.
2. **Riverbank** — the land alongside a river.

The sentence contains two important contextual clues:

- **“by the river”**
- **“flooded”**

A financial institution does not normally represent the object that floods in this context. A **riverbank**, however, can naturally flood.

Therefore:

$$\boxed{Bank = Riverbank}$$

Constraint analysis

Constraint	Financial Bank	Riverbank
Located by a river	Possible	Highly suitable
Can be flooded	Less suitable	Highly suitable

Constraint	Financial Bank	Riverbank
Physical geographical entity	No	Yes
Fits overall sentence	No/weak	**Yes**

Thus, the semantic constraints eliminate the financial meaning and select **riverbank**.

b) Constraint-based disambiguation process

The system can use the following process:

```
         Input Sentence
              ↓
        Identify Ambiguous Word
              ↓
           "bank"
              ↓
       Generate Candidate Senses
          ↙         ↘
   Financial Bank      Riverbank
          ↓           ↓
      Apply Constraints
          ↓
   ┌──────────────┴──────────────┐
   ↓           ↓
"by the river"   "flooded"
   ↓           ↓
 Weak fit     Strong fit
   └──────────────┬──────────────┘
          ↓
      Select Riverbank
```

This demonstrates how **contextual constraints** help a system choose the correct word sense.

c) Represent the meaning using First-Order Predicate Logic

Let the following predicates be defined:

- RiverBank(x) → x is a riverbank.
- River(y) → y is a river.
- By(x,y) → x is located beside y.
- Flooded(x) → x was flooded.
- Storm(s) → s is a storm.
- After(e1,e2) → event $e1$occurred after event $e2$.
- Saved(x,a) → x was saved by action a.
- QuickAction(a) → a was a quick action.

The main semantic relationships can be represented as:

$$RiverBank(x)$$
$$River(y)$$
$$By(x,y)$$

$$
\begin{aligned}
&Flooded(x)\\
&Storm(s)\\
&After(F, loodingStorm)\\
&QuickAction(a)\\
&Saved(x, a)
\end{aligned}
$$

Combined representation:

$$\boxed{RiverBank(x) \wedge River(y) \wedge By(x, y) \wedge Flooded(x) \wedge After(F, loodingStorm) \wedge Saved(x, a) \wedge QuickAction}$$

This representation captures the important entities, events and relationships from the sentence.

d) Convert into an unambiguous target-language sentence

After resolving **bank** as **riverbank**, the sentence can be rewritten as:

“The riverbank flooded after the storm, but quick action saved it.”

This removes the ambiguity completely.

Semantic representation

```
Riverbank
 │
 ├── flooded ──→ after the storm
 │
 └── saved by ──→ quick action
```

The pronoun **“it”** refers to the previously identified **riverbank**.

Therefore:

$$it \rightarrow riverbank$$

e) RST-style discourse structure

The word **“but”** connects two contrasting situations:

1. The riverbank flooded.
2. Quick action saved it.

Therefore, the relationship is **Contrast**.

```
   ROOT
    |
 CONTRAST
 /      \
/        \
```

Situation 1 Situation 2
↓ ↓
Riverbank flooded after the storm | It was saved by quick action

Explanation

The first clause describes a **negative event**:

The riverbank flooded.

The second clause presents an **opposing positive outcome**:

Quick action saved it.

Hence, **Contrast** is the appropriate discourse relation.

f) Advantages of constraint-based WSD

A constraint-based approach provides several benefits over using only statistical prediction.

1. Better ambiguity resolution

It considers semantic and contextual restrictions before selecting a meaning.

2. Improved accuracy

The clues **"river"** and **"flooded"** strongly support the riverbank interpretation.

3. Explainability

The system can explain why it selected a particular sense:

"Bank" was interpreted as **riverbank** because it occurs near "river" and is described as being flooded.

4. Semantic consistency

Predicate logic maintains relationships such as:

$$RiverBank \rightarrow Flooded$$

and

$$RiverBank \rightarrow Saved$$

5. Useful for domain-specific NLP

Constraint-based methods are especially useful in:

- Healthcare
- Banking
- Legal systems
- Customer support
- Search engines

where an incorrect interpretation can lead to an incorrect decision.

Conclusion

Constraint-based Word Sense Disambiguation resolves the ambiguity of **“bank”** by applying contextual and semantic constraints. In this sentence, **“by the river”** and **“flooded”** strongly indicate that *bank* means **riverbank**. Predicate logic can then represent the entities and relationships formally, while RST identifies the **Contrast** relationship between flooding and successful rescue. This combination provides an interpretable and reliable approach to semantic analysis.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
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Date: ____________________	Date: ____________________	Date: ____________________